Q1 An electron and a proton are at distance of 1Å. The moment of this dipole will be (C × m)
(A) 1.6 $\times 10^{19}$
(B) 1.6 $\times 10^{-29}$
(C) 3.2 $\times 10^{19}$
(D) 3.2 $\times 10^{29}$

Q2 The electric field due to a dipole at a distance r on its axis is
(A) Directly proportional to r^3
(B) Inversely proportional to r^3
(C) Directly proportional to r^2
(D) Inversely proportional to r^2

Q3 A given charge is situated at a certain distance from an electric dipole in the end-on position experiences a force F. If the distance of the charge is doubled, the force acting on the charge will be
(A) 2F (B) F/2
(C) F/4 (D) F/8

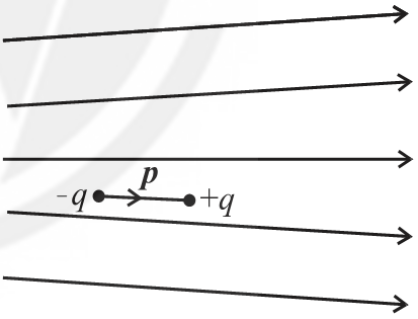
Q4 The electric field at a point on equatorial line of a dipole and direction of the dipole moment
(A) Will be parallel
(B) Will be in opposite direction
(C) Will be perpendicular
(D) Are not related

Q5 When an electric dipole is placed in a uniform electric field, it experiences:
(A) A net force, but no torque
(B) No net force, but a torque
(C) A net force and a torque
(D) Neither a force nor a torque

Q6 The torque τ acting on a dipole in a uniform electric field E is maximum when the angle θ between the dipole moment P and the electric field E is:
(A) 0° (B) 45°
(C) 90° (D) 180°

Q7 **Assertion (A):** The electric field due to an electric dipole is zero at the midpoint between the charges when the dipole moment is placed in a uniform electric field.
Reason (R): In a uniform electric field, the forces on the individual charges of the dipole are equal and opposite, which cancels out the net force on the dipole.
(A) Both Assertion and Reason are correct, and the Reason is the correct explanation of the Assertion.
(B) Both Assertion and Reason are correct, but the Reason is not the correct explanation of the Assertion.
(C) Assertion is correct, but Reason is incorrect.
(D) Assertion is incorrect, but Reason is correct

Q8 Figure shows electric field lines in which an electric dipole *p* is placed as shown in figure. Which of the following statements is correct?



(A) The dipole will not experience any force.
(B) The dipole will experience a force towards right.
(C) The dipole will experience a force upwards.
(D) The dipole will experience a force towards left.

Q9 An electric dipole is placed at an angle of 30° with an electric field intensity 2 $\times 10^5$ N/C. It experiences a torque equal to 4 Nm. The charge on the dipole, if the dipole length is 2 cm, is;

(A) 8 mC (B) 2 mC
(C) 5 mC (D) 7μC

Q10 The electric field due to an electric dipole at a distance r from its centre in axial position is E. If the dipole is rotated through an angle of 90° about its perpendicular axis, the electric field at the same point will be

(A) E
(B) $\frac{E}{4}$
(C) $\frac{E}{2}$
(D) $2E$

Answer Key

Q1	B
Q2	B
Q3	D
Q4	B
Q5	B
Q6	C
Q7	C
Q8	D
Q9	B
Q10	C

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

$p = q \times (2l) = 1.6 \times 10^{-19} \times 10^{-10} = 1.6 \times 10^{-29} C - m.$

Video Solution:

Q2 Text Solution:

$E = \frac{1}{4\pi\varepsilon_0}\frac{2p}{r^3}$

Video Solution:

Q3 Text Solution:

$\text{Force on change } F = q(E_a) = q \times \frac{\frac{k.2p}{r^3}}{\Rightarrow} F$

$\propto \frac{1}{r^3}$

$\text{When } r \rightarrow \text{ doubled; } F \rightarrow \frac{1}{8} \text{ times.}$

Video Solution:

Q4 Text Solution:

The direction of electric field at equatorial point A or B will be in opposite direction, as that of direction of dipole moment.

Video Solution:

Q5 Text Solution:

In a uniform electric field, there is no net force acting on the dipole because the forces on the positive and negative charges cancel each other out. However, the dipole experiences a torque, which tends to align the dipole with the direction of the electric field. The torque is given by:

$T = pE \sin\theta$

Where p is the dipole moment, E is the electric field streghth, and θ is the angle between the dipole moment and the electric field.

Video Solution:

Q6 Text Solution:

The torque experienced by a dipole in a uniform electric field is given by:

$\tau = pE \sin\theta$

The torque is maximum when $\sin\theta = 1$, whenc occurs when $\theta = 90^\circ$, i.e. when the dipole moment is perpendicular to the electric field.

Video Solution:

Q7 **Text Solution:**

The electric field due to a dipole is zero at the midpoint between the charges on the dipole, but this occurs not because of the uniform electric field. It happens due to the symmetry of the dipole itself.

In a uniform electric field, while the forces on the charges are equal and opposite, the net torque is what causes the dipole to align with the field. The assertion is true, but the reason is not relevant to the situation of the field between the dipole charges.

Video Solution:



Q8 **Video Solution:**



Q9 **Text Solution:**

For a dipole

$$\tau = pE\sin\theta, \qquad p = q\ell$$

Given

$\tau = 4\ \mathrm{N\,m},\ E = 2\times10^{5}\ \mathrm{N\,C^{-1}},\ \theta = 30^{\circ},\ \ell = 2\ \mathrm{cm} = 0.02\ \mathrm{m}$

$$p = \frac{\tau}{E\sin\theta} = \frac{4}{(2\times10^{5})(0.5)} = 4\times10^{-5}\ \mathrm{C\,m}$$

$$q = \frac{p}{\ell} = \frac{4\times10^{-5}}{0.02} = 2\times10^{-3}\ \mathrm{C} = 2\ \mathrm{mC}$$

Video Solution:



Q10 **Text Solution:**

For a dipole of moment p at distance r:

- **Axial field:** $E_{\parallel} = \dfrac{1}{4\pi\varepsilon_0}\dfrac{2p}{r^3} = E.$
- **Equatorial field (after 90° rotation):**

$$E_{\perp} = \frac{1}{4\pi\varepsilon_0}\frac{p}{r^3} = \tfrac{1}{2}E_{\parallel} = \frac{E}{2}.$$

Video Solution:

